Airspace Unlimited Interview

Audio transcript

Doug Meyerhoff's my name. I'm an air traffic controller by trade, originally born in Canada but now living in the United Kingdom. Airspace Unlimited Scotland grew out of a conversation between myself and a business partner when he came to me with a problem regarding the efficient use of airspace reservations from the military, and the impact it had on civilian traffic. Oftentimes the airspace reservations are block-booked with a rather, sort of, crude methodology, and the use of those airspace reservations is quite disadvantageous to the airlines because they are often avoiding airspace which really doesn't need to be avoided. We got together and we discussed how can we best improve the efficiency of the airspace reservation system, but then also by doing that, decrease the cost to the airlines, increase the efficiency of the military operation, collectively reducing the overall CO_2 emissions of the aviation industry.

As you can imagine, an aircraft does not want to fly directly into the wind. He wants to take advantage of the tailwinds and so what we're able to do with the predictive winds aloft forecasts that are available now, is that we can see where the airlines want to go and how it's going to impact. If you think of the North Atlantic, oftentimes you'll see traffic coming on a quite northerly route from North America to Europe, and then, if the jet stream moves, that sort of traffic demand moves with the jet stream. And the idea is that by being able to predict where the civilian traffic wants to be, we can then go to the military mission planners and say, well, what are you trying to accomplish? Well, we're in the middle of this training program. OK, well, does it need to actually happen north-south? Can we turn it and make it go east-west? Thereby, if you've got an elongated rectangle of airspace that they require, oriented north-south, that obviously creates a bigger impact on the civilian traffic having to go around. Whereas if we can reorient it east-west, your profile is then reduced considerably, allowing a more efficient and more direct route for the commercial operators.

As the ability to accurately predict the weather forecast, and I'm talking specifically of the winds aloft, that has improved the efficiency of the airlines and the aircraft specifically, but the system has not followed along with that efficiency gain. So we're still working on a very rigid, geostationary airspace concept whereby we now can see the winds very accurately 72 hours in advance, and by using those predictive winds, we can then look at where is the best and most efficient place to place these reservations so that it has a minimal impact on the commercial aviation, again, recognising that the military have an operational role that they need to fulfill. But we can then improve the relationship between the two, thereby reducing the overall cost and the overall climate impact. By being able to fine-tune the exact demands of the commercial operators on the military operators, we can then shape the actual air space dimensions and the reservations themselves rather than being relatively crude rectangular reservations, where if you imagine in an aircraft, he's never going to go right into the corner. We can shave those corners, we can shape the airspace reservations to exactly what's required. Again, giving the military precisely what they require as far as airspace in which to operate and to train, but then again, reducing the amount of wasted airspace.

From the commercial perspective, being able to predict it in advance means that we can reorient and we can optimise the airspace before they get to their flight planning stage. Oftentimes, airlines will flight plan their operations, for instance, I'll give you an example in Europe for the short haul carriers easyJet, Ryanair, British Airways, they're sort of their European operations, they tend to block book their flight plans 12 hours in advance. That 12-hour window then locks them into those flight plans. And if there's an opportunity to optimise the airspace within that 12-hour window, the airlines are often unable to take up the advantages that are presented. So by being able to predict more than 12 hours, like I said up to 72 hours in advance, we can then get that efficiency built into the system before the airlines actually follow their flight plans. Thereby they're getting the efficient air space without having to do any sort of reverse engineering of their flight plans, which is oftentimes difficult for them with their staffing and system requirements. The weather data that we would use specifically is the winds aloft forecasts. As the air masses move and the high pressure and low pressure systems move, that creates the waves and creates the jet stream effect and so on. And as you increase in altitude, the winds will change direction and velocity. So being able to have accurate winds aloft forecasts really helps us fine-tune our predictive ability so we try to use the UK Met Office. There's an ensemble forecast system in Europe which is proving to be extremely effective for us. And then there's the European Medium Range Weather Forecast office as well, which they've been extremely helpful and quite supportive of us as well.

We want to get the data from as close to the original source as possible. That gives us the ability to then feed that information directly into our algorithm, where our internal system then takes the raw data and produces it and applies it to our algorithm, giving us the most accurate forecasts and track prediction.

Our business currently is focused on the military reservations, and so we see the military authority in whichever country as probably the primary source of, or the primary target audience, for our service. However, there is an application whereby we could go to the air navigation service provider so that would be the air traffic control people. In the UK it would be NATS, Eurocontrol or whomever the national authority is. We see that there's a broader footprint there. We also see potential for looking at the maritime environment by applying the routing algorithms that we have from the airlines but then applying it to ships as well whereby we can reduce their track mileage and we can improve their predictability when it comes to actual fuel burn and so on. Our main thrust at this stage is the environmental aspect of that, although I think increasingly, particularly with the events in the recent couple of months in the Ukraine, although the environmental aspect is still quite important, the more critical and the more pressing, you know, flashing light as it were, is the cost of fuel. I think the practical cost of fuel and I guess the availability of fuel as well. But the environmental one is sort of the underlying current which is running through our offering.